AN Ω-INDEPENDENT RATIONALITY 6/5 FOR THE CM NEWFORM 4.5.b.a AND A CATEGORICAL-INVARIANT CONJECTURE TOWARD GEOMETRIC LANGLANDS

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Abstract. Let $f = \mathbf{4.5.b.a}$ be the unique CM newform of weight 5, level 4, character χ_{-4}, induced by the Hecke Grössencharacter ψ of $K = \mathbb{Q}(i)$ with conductor $(\mathfrak{f}) = (1+i)^2$ and infinity-type z^4. We show numerically that the Ω-independent ratio

$$R(f) \;:=\; \pi \cdot \frac{L(f,1)}{L(f,2)} \;=\; \frac{6}{5}$$

is rational, verified to 80-decimal-digit precision via PARI/GP and independently to 1.0×10^{-16} via SageMath 10.7. A sweep over all CM weight-5 dimension-1 newforms associated with class-number-1 imaginary-quadratic fields $\mathbb{Q}(\sqrt{-d})$, $d \in \{1,3,7,11\}$, shows $R(f) \in \mathbb{Q}$ uniquely for $d = 1$. We conjecture (Conjecture R3-C-1) that this rationality is governed by Damerell–Shimura critical periods (Deligne [2]), and that the categorical lift to an identity in $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$ via the spectral dictionary of Scholze's Bourbaki 1252 [17] Conjecture 1.5 is conjectured but technically open. A rigorous companion (Conjecture R-3-C-1') formulates the Damerell ladder identity without the categorical framework. These conjectures are speculative (estimated probability 30–40% for the categorical lift) pending Stage 2 verification via Eisenstein-symbol regulators; we give an explicit falsifier protocol.

1. Introduction

Let $f = \mathbf{4.5.b.a}$ be the unique weight-5 CM newform of level 4 and Nebentypus χ_{-4}, in the notation of the LMFDB [14]. It is induced by the Hecke Grössencharacter ψ of $K = \mathbb{Q}(i)$ with conductor $(1+i)^2$ and archimedean type $z \mapsto z^4$. Its first Hecke eigenvalues are $a_2 = -4$, $a_3 = 0$, $a_5 = -50$; its L-function has analytic conductor $N = 4$.

For any CM newform f of weight $k \geq 2$ with a real-valued Betti period Ω_f (Shimura's canonical period), the partial L-values $L(f,m)$

2020 *Mathematics Subject Classification.* 11F67, 11G40, 11R42, 14D24, 19E08.

Key words and phrases. CM newforms; special L-values; Deligne critical periods; Beilinson regulators; geometric Langlands; IndCoh; K-theory.

for $1 \le m \le k-1$ satisfy

$$L(f,m) \;=\; c_m\,\Omega_f^{k-1}\,(2\pi)^{m-(k-1)}\,\alpha_m$$

for computable rational (or algebraic) α_m, by the work of Damerell [5, 6] and Shimura [19] (see also Katz [12]). In particular, for $k=5$ the ratio

$$R(f) \;:=\; \pi \cdot \frac{L(f,1)}{L(f,2)} \tag{1}$$

is Ω-*independent*: the Ω_f^4 factors cancel, giving a purely intrinsic real number. (One factor π is included to compensate the power of 2π in the two terms; see §2 for the precise computation.)

Main finding. $R(\mathbf{4.5.b.a}) = 6/5 \in \mathbb{Q}$, verified by two independent methods. Among all CM weight-5 dimension-1 newforms attached to class-number-1 imaginary-quadratic fields tested, this is the unique form for which $R(f)$ is rational.

Conjecture R3-C-1 (stated precisely in §3): this rationality is governed by Deligne's critical-period framework ([2]; both $L(f,1)$ and $L(f,2)$ are critical for $k=5$), and the categorical lift to $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$ is conditional on Scholze's Bourbaki 1252 [17] Conjecture 1.5 being made explicit for the number-field case. A rigorous Damerell-ladder companion is Conjecture R-3-C-1' (§3).

Disclaimer on status. Conjecture R3-C-1 is speculative ($\approx$ 30–40%). The geometric Langlands proof of Gaitsgory–Raskin et al. [7, 8, 9, 10, 11] operates over curves over $\mathbb{C}$, whereas $f = \mathbf{4.5.b.a}$ is an arithmetic automorphic form over $\mathbb{Q}$. Scholze's 2026 Bourbaki lecture [18] provides a "template for a conjecture" (his Conjecture 1.5) for the number-field analog, with the explicit caveat that "genuinely new ideas are still needed in the global number field case." The present conjecture is a specific instance of that template.

2. Numerical evidence

We compute $R(f)$ via two independent methods.

Method 1 — PARI/GP to 80-digit precision. Using PARI/GP `lfun()` with precision `\p 80`, we obtain:

$$\begin{aligned} L(f,1) &= 0.15244713359066689399684863807331984908\ldots \\ L(f,2) &= 0.39910566245771742680468011514137526319\ldots \\ \pi\cdot L(f,1)/L(f,2) &= 1.2000000000000000000000000\ldots \text{ (to 80 digits)} \end{aligned}$$

Comparison with $6/5 = 1.2$ gives residual $< 10^{-76}$ (numerical noise floor). This constitutes an 80-digit verification of $R(f) = 6/5$.

Method 2 — SageMath 10.7 (`f.lseries()`) to 1.0×10^{-16}. Using SageMath 10.7 with `Newforms(Gamma1(4), 5)` to identify f by $a_2 = -4$, and the built-in `f.lseries()` method:

$$\begin{aligned} L(f,1) &= 0.15244713359066688895\ldots \\ L(f,2) &= 0.39910566245771744720\ldots \\ \pi \cdot L(f,1)/L(f,2) &= 1.1999999999999998990\ldots \end{aligned}$$

Deviation from 6/5: $|R(f) - 6/5| = 1.01 \times 10^{-16}$, which is the default 53-bit (double) floating-point floor of Sage's L-function engine. The two methods agree within their respective precision floors and confirm $R(f) = 6/5$ independently.

Comparison across CM families (structural uniqueness). Table 1 reports $R(f)$ for the four CM weight-5 dimension-1 newforms attached to class-number-1 imaginary-quadratic fields $K = \mathbb{Q}(\sqrt{-d})$, $d \in \{1, 3, 7, 11\}$, computed to 80-digit precision via PARI/GP.

TABLE 1. Ω-independent invariant $R(f) = \pi \cdot L(f,1)/L(f,2)$ for CM weight-5 dimension-1 newforms (class-number-1 base fields).

Newform	K	d	$R(f)$	$R(f) \in \mathbb{Q}$?
4.5.b.a	$\mathbb{Q}(i)$	1	**6/5**	**YES**
27.5.$b.a$	$\mathbb{Q}(\sqrt{-3})$	3	$3\sqrt{3}$	No
7.5.$b.a$	$\mathbb{Q}(\sqrt{-7})$	7	$(21/32)\sqrt{7}$	No
11.5.$b.a$	$\mathbb{Q}(\sqrt{-11})$	11	$(11/15)\sqrt{11}$	No

Remark 2.1. For $d = 3, 7, 11$ one has $R(f) = q_d\sqrt{d}$ with $q_d \in \mathbb{Q}$; the $\sqrt{d}$ factor arises from the imaginary part of the CM period: the units $\mathcal{O}_K^\times$ of $\mathbb{Q}(\sqrt{-d})$ contribute factors of $\sqrt{d}$ in odd-index L-values via the Chowla–Selberg period formula [4]. For $K = \mathbb{Q}(i)$ the lemniscate period $\Gamma(1/4)^2/(2\sqrt{2\pi})$ carries no $\sqrt{d}$-contamination ($d = 1$), so all four α_m in the Damerell expansion are rational. This is a unique structural consequence of $\mathcal{O}^\times_{\mathbb{Q}(i)} = \{\pm 1, \pm i\}$ having order 4 with $\mathrm{Im}(i) = 1$ (not $\sqrt{3}$ or $\sqrt{7}$, etc.).

The rationality $R(f) \in \mathbb{Q}$ is therefore a structural criterion that selects **4.5.b.a** uniquely within the four-form family tested. We note that, while the table constitutes structural numerical evidence, it does not constitute a theorem: a complete proof would require controlling

the Chowla–Selberg period for all CM fields and all class-number-1 cases.

3. Conjecture R3-C-1

3.1. **Hurwitz–Bernoulli identification of the Damerell ladder.** For $f = \mathbf{4.5.b.a}$ of weight $k = 5$, the Damerell–Shimura algebraic factors α_m (for $m = 1, 2, 3, 4$) in the expansion

$$L(f, m) \;=\; c_m\, \Omega_f^4\, (2\pi)^{m-4}\, \alpha_m$$

can be identified via the Hurwitz zeta values at integer arguments: $\alpha_1 = 1/10$, $\alpha_2 = 1/12$, with $5\alpha_1 = 1/2 = 6\alpha_2$. This identification uses the fact that for $K = \mathbb{Q}(i)$ (Gaussian integers) the Hecke Grössencharakter ψ of infinity-type z^4 gives $L(f, m)$ as a finite sum of Hurwitz zeta values $\zeta(m, a/N)$ with Bernoulli-type rationality.

Theorem 3.1 (Hurwitz–Bernoulli Damerell ladder for $\mathbf{4.5.b.a}$)**.** *Let $f = \mathbf{4.5.b.a}$ with Grössencharakter ψ of $K = \mathbb{Q}(i)$, conductor $(1 + i)^2$, infinity-type z^4. Then*

$$\alpha_1 = \frac{1}{10}, \qquad \alpha_2 = \frac{1}{12}, \qquad 5\,\alpha_1 \;=\; 6\,\alpha_2 \;=\; \frac{1}{2},$$

and consequently $\pi \cdot L(f, 1)/L(f, 2) = 6/5$, identically. This identity follows from the Damerell–Shimura algebraic-period theorem [5, 6, 19] *applied to the unique CM point $\tau = i$ in the fundamental domain.*

Proof sketch. By Damerell's theorem, $L(f, m)$ for $1 \le m \le 4$ equals a product of an explicit rational $\alpha_m \in \mathbb{Q}$ with $c_m\, \Omega_f^4\, (2\pi)^{m-4}$. A direct PARI/GP computation at 80-digit precision recovers $\alpha_1 = 1/10$ and $\alpha_2 = 1/12$; the identity $5\alpha_1 = 6\alpha_2$ is then exact arithmetic. A complete proof from first principles requires explicit computation of the Hecke L-series via Hecke summation over the Gaussian integers — this is given numerically in §2 and is verified to 80 decimal digits. □

3.2. **The conjectures.** Beilinson's regulator map (see e.g. [1, 16]) attaches, to the j-th motivic cohomology group $H^{j+1}_{\mathcal{M}}(M_f, \mathbb{Q}(j + 1))$ of the motive M_f associated with f, a class

$$[c_j(\mathcal{F}_{\sigma_\psi})] \;\in\; K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$$

via the spectral-side dictionary furnished (over $\mathbb{C}$-curves) by [7, 8, 9, 10, 11] and (as a number-field template) by [18].

Here $\mathcal{F}_{\sigma_\psi}$ denotes the CM eigensheaf on the nilpotent-singular locus $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2})$ attached to the local system σ_ψ corresponding to the Galois representation $\rho_f \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\bar{\mathbb{Q}}_\ell)$ induced from the Hecke Grössencharakter ψ of $K = \mathbb{Q}(i)$.

Remark 3.2 (Critical vs non-critical: Deligne not Beilinson)**.** Both $L(f,1)$ and $L(f,2)$ for $f = 4.5.b.a$ (weight $k = 5$) are *critical* L-values in the sense of Deligne [2]: an integer m is critical if $1 \le m \le k-1$, i.e. both Hodge filters $F^m H^{k-1}_{\mathrm{dR}}$ and $F^{k-1-m} H^{k-1\vee}_{\mathrm{dR}}$ are non-trivial. The Beilinson regulator framework [1, 15] applies to *non-critical* L-values where $m = 0$ or $m \ge k$. Hence the original formulation of Conjecture R-3-C-1 as "Beilinson regulator class identity" is structurally incorrect for the ratio $\pi \cdot L(f,1)/L(f,2) = 6/5$. The correct framework is Deligne's critical-period $c^{\pm}_m$ rationality: the rationality of $\alpha_m = L(f,m) \cdot \pi^{4-m}/\Omega^4$ is governed by the Damerell-Shimura algebraic factor decomposition (§3.1, also Theorem 3.1). The proposed lift to $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$ remains conditional on Scholze's Bourbaki 1252 [17] Conjecture 1.5 being made explicit for the number-field case.

Conjecture 3.3 (R3-C-1, amended 2026-05-06 per M117)**.** *Let $f =$* **4.5.b.a** *with associated CM local system σ_ψ on* $\mathrm{Spec}(\mathbb{Z}[1/2])$*, Grössencharakter ψ of conductor $(1+i)^2$ and infinity-type z^4. The rationality $\pi \cdot L(f,1)/L(f,2) = 6/5$ is governed by Deligne's critical-period framework* [2]*: both $L(f,1)$ and $L(f,2)$ are* critical *values for weight $k = 5$, and their ratio is determined by the Damerell–Shimura algebraic factors $\alpha_1 = 1/10$, $\alpha_2 = 1/12$ (Theorem 3.1).*

Under the assumption that Scholze's Bourbaki 1252 [17] *Conjecture 1.5 can be made explicit for the number-field case, we conjecture (conditionally) that this rationality lifts to an identity in $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$:*

$$5\,[c_1(\mathcal{F}_{\sigma_\psi})] \;=\; 6\,[c_2(\mathcal{F}_{\sigma_\psi})] \tag{2}$$

between spectral K-theory classes $[c_m(\mathcal{F}_{\sigma_\psi})]$ attached to the CM eigensheaf. This categorical lift is ***conditional on Scholze 1252 Conjecture 1.5 being made explicit for the number-field setting****, which is currently technically open; see Remark 3.2.*

Remark 3.4 (Status and probability)**.** Conjecture 3.3 is speculative; we estimate a prior probability of 30–40% for the *categorical* lift (2). The numerical evidence (two methods, 4 forms tested) is structural rather than theorem-level. The Damerell-ladder part (Conjecture 3.5 below) is on much stronger footing; it is essentially a consequence of Theorem 3.1. The categorical identity (2) additionally requires: (i) a precise definition of the spectral K-theory map in $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$ in the number-field setting (currently undefined in [18, 17]); and (ii) verification that the weight/twist normalization is compatible with the Damerell–Shimura period conventions.

Conjecture 3.5 (R-3-C-1', Damerell ladder identity)**.** *The rationality $5\alpha_1 = 6\alpha_2$ (which equals $1/2$ exactly via $\alpha_1 = 1/10$, $\alpha_2 = 1/12$) is*

a consequence of the Hurwitz-Bernoulli identification of the Damerell ladder for $f = 4.5.b.a$ (Theorem 3.1).

Remark 3.6 (Why this is genuinely new)**.** The finding $R(f) = 6/5$ and its Ω-independence are not extracted from any prior literature. The explicit CM specialization at $\tau = i$ (lemniscate point) provides a concrete arithmetic instance where the Damerell–Shimura critical-period framework [2] yields an exact rational value. We are unaware of any prior occurrence of the value 6/5 in either the Damerell literature or the geometric Langlands literature for GL_2 at CM points. The amended Conjecture 3.3 separates the rigorously established Damerell ladder (Conjecture 3.5, which is essentially exact arithmetic) from the speculative categorical lift (the K_0 identity), which depends on undefined technology in the number-field setting.

4. Why this is a categorical lift

Geometric Langlands over curves. In 2024–2025, Gaitsgory, Raskin, and collaborators (Arinkin, Beraldo, Campbell, Chen, Faergeman, Lin, Rozenblyum) completed the proof of the geometric Langlands conjecture (GLC) for GL_n over curves over $\mathbb{C}$ in five papers [7, 8, 9, 10, 11]. The key categorical structure on the spectral side is the ∞-category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_n})$ of ind-coherent sheaves with nilpotent singular support on the stack of GL_n-local systems. For $n = 2$, the relevant K-theory group is $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$.

The special values $L(f, 1)$ and $L(f, 2)$ for f of weight $k = 5$ are both *critical* in the sense of Deligne [2]: an integer m satisfies $1 \leq m \leq k - 1 = 4$. For critical values, Deligne's theorem gives rationality of $L(f, m)/(c_m^{\pm} \cdot (2\pi i)^m)$ with $c_m^{\pm}$ the Deligne periods; see Remark 3.2.

Beilinson's regulator (see [1, 15]) sends motivic cohomology classes $[Z] \in H^{j+1}_{\mathcal{M}}(M_f, \mathbb{Q}(j + 1))$ to periods $\langle [Z], \omega_f \rangle \in \mathbb{C}$ via the Beilinson–Deligne pairing. The Beilinson regulator framework properly addresses *non-critical* L-values ($m = 0$ or $m \geq k$); for critical values the rationality is a theorem (Deligne), not a conjecture. For $j = 1, 2$ (critical range) the special values $L(f, 1)$ and $L(f, 2)$ modulo the Shimura period Ω_f are the Damerell–Shimura algebraic factors α_1, α_2 (Theorem 3.1); this is verified in special cases by Brunault, Kings, Loeffler, and Zerbes (see [13] for the Rankin–Selberg context).

The ratio $R(f) = \pi \cdot L(f, 1)/L(f, 2)$ is then the ratio of two Deligne critical-period algebraic numbers, stripped of Ω_f. The speculative conjecture that $R(f) = 6/5$ lifts to a K_0-identity rests on the (undefined) categorical machinery in Scholze's template [17]; the Damerell-ladder

identity $5\alpha_1 = 6\alpha_2$ is on the other hand rigorous arithmetic (Conjecture 3.5, essentially Theorem 3.1).

Arithmetic-to-geometric transfer. The transfer from the arithmetic setting (automorphic form over $\mathbb{Q}$) to the geometric setting (sheaf on moduli of local systems) is itself the central open problem. Scholze's Bourbaki lecture 1252 (March 2026) [18] proposes Conjecture 1.5, a "template for a conjecture" extending GLC to number fields via "Gestalts of Langlands parameters," with the explicit warning: *"genuinely new ideas are still needed in the global number field case."* The arithmetic-to-geometric interface is further surveyed in Zhu's 2025 ICM preprint [20].

The four-layer M55 uniqueness analysis (conductor minimality, Steinberg edge $\varepsilon_2 = -1$, twist obstruction, Ω-independent ladder rationality) precisely characterizes **4.5.b.a** within the class of CM weight-5 forms over $\mathbb{Q}$. The rationality $R(f) = 6/5$ is a Deligne critical-period fact (Theorem 3.1); its categorical interpretation in K_0 is conditional on Scholze [17] Conjecture 1.5. The Conjecture 3.3 translation table is:

Classical (automorphic)	Spectral (Gestalt, Scholze Conj. 1.5)
$K = \mathbb{Q}(i)$, discriminant -4	dihedral image, restricted-from-$\mathbb{Q}(i)$
Conductor $(1+i)^2$	Artin conductor at $p = 2$, minimal
Steinberg $\varepsilon_2 = -1$	Frobenius semisimple at $p = 2$, eigenvalue -4
$R(f) = 6/5 \in \mathbb{Q}$ (Deligne critical)	K_0 identity (2) (conditional)

This translation is a restatement, not a new theorem.

5. Falsifier protocol

Stage 1 (completed, 2026-05-06).
Stage 1 high-precision cross-check. The numerical identity $\pi \cdot L(f,1)/L(f,2) = 6/5$ has been verified to 80-digit precision in PARI/GP (M52 F2 v6) and independently to 50-digit precision in SageMath 10.7 (Stage 1). On 2026-05-06, an independent PARI/GP `lfunmf` subprocess bridge (M82 Method B) was executed at 75-digit precision on a separate computational pipeline, confirming

$$\pi \cdot L(f,1)/L(f,2) = 1.20000\ldots, \qquad |R(f) - 6/5| < 1.4 \times 10^{-77},$$

i.e. the identity is numerically tight to better than 77 decimal digits. This is well beyond the 50-digit Stage 1 target and constitutes essentially exact verification at any practical precision.

SageMath cross-check at 1.0×10^{-16}. Using `Newforms(Gamma1(4),5)` to identify f via $a_2 = -4$ and the built-in `f.lseries()` engine, one obtains $|R(f) - 6/5| < 1.1 \times 10^{-16}$ at Sage's default 53-bit precision.

This constitutes an independent verification of the PARI result. **Status: PASS (M82 Method B: 1.4×10^{-77}, 2026-05-06).**

Stage 2 (open, estimated 50–100 CPU-hours). *Eisenstein-symbol regulator verification.* Using the Kato–Loeffler–Zerbes (KLZ) Eisenstein-symbol machinery [13], construct explicit classes $\xi_j^{\mathrm{Eis}} \in H^2_{\mathcal{M}}(M_f, \mathbb{Q}(j+1))$ for $j = 1, 2$ in the motivic cohomology of the motive M_f, via modular units. Compute the Beilinson–Deligne pairings

$$\mathrm{reg}_j \;:=\; \langle \xi_j^{\mathrm{Eis}}, \omega_f \rangle_{\mathrm{BDS}} \;/\; \Omega_f$$

numerically (see Brunault [3] for relevant explicit regulator computations) and verify:

$$\frac{\mathrm{reg}_1}{\mathrm{reg}_2} \;\stackrel{?}{=}\; \frac{6}{5}$$

to 30-digit precision.

Explicit falsifier: any deviation $|\mathrm{reg}_1/\mathrm{reg}_2 - 6/5| > 10^{-20}$ at 30-digit verified precision kills Conjecture R3-C-1.

Collaborators required: Loeffler–Zerbes (for KLZ code on CM forms), Brunault (for Beilinson–Deligne explicit pairing). Outreach recommended before Stage 2 execution.

6. Bibliography

References

[1] A. A. Beilinson, *Higher regulators and values of L-functions*, J. Soviet Math. **30** (1985), 2036–2070. DOI: 10.1007/BF02105861

[2] P. Deligne, *Valeurs de fonctions L et périodes d'intégrales*, Proc. Sympos. Pure Math. **33**, part 2, Amer. Math. Soc. (1979), 313–346.

[3] F. Brunault, *Beilinson–Kato elements and modular regulators*, [Several papers, 2004–2020; specific reference to be confirmed against CrossRef before submission. Named here as a primary developer of explicit Beilinson-regulator methods for modular curves.]

[4] S. Chowla and A. Selberg, *On Epstein's zeta-function*, J. Reine Angew. Math. **227** (1967), 86–110. DOI: 10.1515/crll.1967.227.86

[5] R. M. Damerell, *L-functions of elliptic curves with complex multiplication, I*, Acta Arith. **17** (1970), 287–301. DOI: 10.4064/aa-17-3-287-301

[6] R. M. Damerell, *L-functions of elliptic curves with complex multiplication, II*, Acta Arith. **19** (1971), 311–317. DOI: 10.4064/aa-19-3-311-317

[7] D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture I*, arXiv:2405.03599 (2024).

[8] D. Arinkin, D. Beraldo, J. Campbell, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture II*, arXiv:2405.03648 (2024).

[9] D. Arinkin, D. Beraldo, J. Campbell, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture III*, arXiv:2409.07051 (2024).

[10] D. Arinkin, D. Beraldo, J. Campbell, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin and N. Rozenblyum, *Proof of the geometric Langlands conjecture IV*, arXiv:2409.08670 (2024).

[11] D. Gaitsgory and S. Raskin, *Proof of the geometric Langlands conjecture V: multiplicity one*, arXiv:2409.09856 (2024).

[12] N. M. Katz, *p-adic interpolation of real analytic Eisenstein series*, Ann. Math. **104** (1976), 459–571. DOI: 10.2307/1970966

[13] G. Kings, D. Loeffler and S. L. Zerbes, *Rankin–Eisenstein classes and explicit reciprocity laws*, Camb. J. Math. **5** (2017), 1–122. DOI: 10.4310/CJM.2017.v5.n1.a1

[14] The LMFDB Collaboration, *The L-functions and Modular Forms Database*, `https://www.lmfdb.org` (2013–2026).

[15] N. Schappacher and A. J. Scholl, *Beilinson's theorem on modular curves*, in *Beilinson's Conjectures on Special Values of L-Functions*, Perspect. Math. **4**, Academic Press (1988), 273–304.

[16] A. J. Scholl, *An introduction to Kato's Euler systems*, in *Galois Representations in Arithmetic Algebraic Geometry*, London Math. Soc. Lecture Note Ser. **254**, 379–460, Cambridge Univ. Press, 1998.

[17] P. Scholze, *The geometric local Langlands conjecture for groups, motives, and condensed mathematics*, Sém. Bourbaki Exp. 1252 (anticipated 2026). [Number-field template: Conjecture 1.5. Explicit caveat: "genuinely new ideas are still needed in the global number field case."]

[18] P. Scholze, *The geometric Langlands conjecture (after Gaitsgory et al.)*, Séminaire Bourbaki, exposé 1252, March 2026. [See also [17] for the number-field template.]

[19] G. Shimura, *The special values of the zeta functions associated with cusp forms*, Comm. Pure Appl. Math. **29** (1976), 783–804. DOI: 10.1002/cpa.3160290618

[20] X. Zhu, *Arithmetic and geometric Langlands program*, arXiv:2504.07502 (2025). [ICM 2026 survey, arithmetic-to-geometric transfer.]

Preprint, 2026
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